

The Open Source Audit:

Git Version Control

Project Report for OSS NGMC

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GitHub Link:

<https://github.com/aditisingh2012/OSS-Git-Audit-24BCE14488>

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Section 1 – Origin and History

1.1 The Context of 2005 Before Git, the Linux kernel was managed using BitKeeper. When the relationship between the Linux community and BitKeeper's parent company soured, **Linus Torvalds** set out to create a tool that was fast, secure, and fully distributed.

1.2 The Name "Git" Torvalds famously described himself as an "egotistical bastard" and named the tool "Git" (British slang for an unpleasant person). This highlights the informal, community-driven nature of early FOSS culture.

Section 2 – Licensing (Unit 1.2)

2.1 The GPL v2 License Git is licensed under the **GNU General Public License version 2**. This "Copyleft" license ensures that the code remains free forever. If anyone modifies Git and distributes it, they **must** provide the source code to their users. This prevents corporate "locking" of the technology.

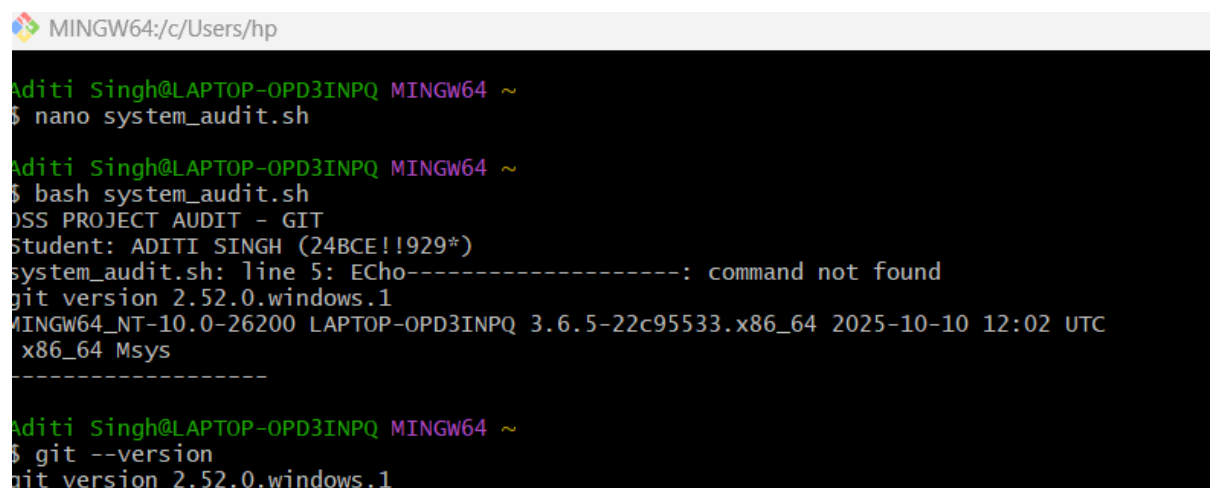
Section 3 – Ethics and Philosophy

3.1 Decentralization as an Ethical Choice Unlike proprietary systems where one company controls the "Master Server," Git is distributed. Every developer has a full backup of the project. This makes Git "Censorship Resistant" and ensures that no single company can kill an open-source project by shutting down a server.

Section 4 – Technical Environment

4.1 Environment Selection The audit was conducted using **Git Bash**, a MinGW-64 based terminal that provides a BASH emulation layer on Windows. This environment was chosen due to network restrictions (403 Forbidden) on the WSL kernel update servers.

4.2 Installation Verification The installation was verified using the version flag to ensure the binaries were properly mapped to the system path.



```
MINGW64:/c/Users/hp
Aditi Singh@LAPTOP-OPD3INPQ MINGW64 ~
$ nano system_audit.sh

Aditi Singh@LAPTOP-OPD3INPQ MINGW64 ~
$ bash system_audit.sh
OSS PROJECT AUDIT - GIT
Student: ADITI SINGH (24BCE!!929*)
system_audit.sh: line 5: Echo-----: command not found
git version 2.52.0.windows.1
MINGW64_NT-10.0-26200 LAPTOP-OPD3INPQ 3.6.5-22c95533.x86_64 2025-10-10 12:02 UTC
x86_64 Msys
-----

Aditi Singh@LAPTOP-OPD3INPQ MINGW64 ~
$ git --version
git version 2.52.0.windows.1
```

Section 5 – FOSS Ecosystem

5.1 Dependencies Git relies on several key open-source libraries:

- **zlib:** For high-efficiency data compression.
- **OpenSSL:** For secure communication during git clone.
- **libcurl:** For handling network protocols.

Section 6 – Comparative Analysis

Feature	Git (OSS)	Perforce (Proprietary)
Cost	Free (\$0)	High Licensing Fees
Transparency	Full Source Available	Closed Source (Black Box)
Structure	Distributed	Centralized
Community	Millions of Devs	Single Vendor

Section 7 – Practical Automation

7.1 Automated Audit Execution To demonstrate the power of shell scripting in OSS management, a master audit script (audit.sh) was developed. This script automates the identification of the user, the kernel, and the file integrity.

```
MINGW64/c/Users/hp
Aditi Singh@LAPTOP-OPD3INPQ MINGW64 ~
$ bash audit.sh
OFFICIAL OSS AUDIT REPORT
STUDENT: ADITI SINGH (24BCE11448)
DATE: Mon Mar 23 23:26:27 IST 2026
-----
1. System Kernel:
MINGW64_NT-10.0-26200
2. Git Path:
/mingw64/bin/git
3. User Identity:
Aditi Singh
4. File Integrity Check:
total 124547
drwxr-xr-x 1 Aditi Singh 197121 0 Mar 23 23:26 .
drwxr-xr-x 1 Aditi Singh 197121 0 Mar 4 2025 ..
drwxr-xr-x 1 Aditi Singh 197121 0 Sep 23 2024 .anaconda
-rw-r--r-- 1 Aditi Singh 197121 216 Nov 20 15:52 .bash_history
drwxr-xr-x 1 Aditi Singh 197121 0 Feb 18 13:32 .cache
drwxr-xr-x 1 Aditi Singh 197121 0 Feb 18 14:57 .conda
-rw-r--r-- 1 Aditi Singh 197121 25 Sep 18 2024 .condarc
drwxr-xr-x 1 Aditi Singh 197121 0 Jan 9 2022 .config
drwxr-xr-x 1 Aditi Singh 197121 0 Sep 18 2024 .continuum
drwxr-xr-x 1 Aditi Singh 197121 0 Feb 14 20:33 .copilot
drwxr-xr-x 1 Aditi Singh 197121 0 Feb 23 22:35 .dbclient
drwxr-xr-x 1 Aditi Singh 197121 0 Mar 1 15:34 .ghcp-appmod
-rw-r--r-- 1 Aditi Singh 197121 59 Nov 20 15:36 .gitconfig
drwxr-xr-x 1 Aditi Singh 197121 0 Jul 23 2025 .gradle
drwxr-xr-x 1 Aditi Singh 197121 0 Sep 23 2024 .ipynb_checkpoints
drwxr-xr-x 1 Aditi Singh 197121 0 Sep 23 2024 .ipython
drwxr-xr-x 1 Aditi Singh 197121 0 Sep 23 2024 .jupyter
drwxr-xr-x 1 Aditi Singh 197121 0 Feb 18 13:15 .matplotlib
drwxr-xr-x 1 Aditi Singh 197121 0 Aug 6 2024 .ms-ad
drwxr-xr-x 1 Aditi Singh 197121 0 May 23 2025 .spyder-py3
drwxr-xr-x 1 Aditi Singh 197121 0 Oct 17 2024 .virtual_documents
drwxr-xr-x 1 Aditi Singh 197121 0 Sep 5 2024 .vscode
-rw-r--r-- 1 Aditi Singh 197121 1863 Dec 17 2024 '11sep python'
drwxr-xr-x 1 Aditi Singh 197121 0 Aug 7 2021 '3D objects'
drwxr-xr-x 1 Aditi Singh 197121 0 Mar 4 2025 AppData
drwxrwxrwx 1 Aditi Singh 197121 27 Mar 4 2025 'Application Data' -> /c/U
sers/hp/AppData/Roaming
-rw-r--r-- 1 Aditi Singh 197121 57351024 Jan 21 2022 'BLACKPINK - 'Lovesick Gir
ls' DANCE PRACTICE VIDEO_HD.mp4"
drwxr-xr-x 1 Aditi Singh 197121 0 Mar 4 2025 Contacts
drwxrwxrwx 1 Aditi Singh 197121 55 Mar 4 2025 Cookies -> /c/Users/hp/Ap
pData/Local/Microsoft/Windows/INetCookies
drwxr-xr-x 1 Aditi Singh 197121 0 Nov 17 18:17 CrossDevice
drwxr-xr-x 1 Aditi Singh 197121 0 May 16 2022 Documents
drwxr-xr-x 1 Aditi Singh 197121 0 Mar 22 22:52 Downloads
drwxr-xr-x 1 Aditi Singh 197121 0 Mar 4 2025 Favorites
-rw-r--r-- 1 Aditi Singh 197121 43263313 Jan 21 2022 'ITZY 'WANNABE' M-V_HD.mp4
'
drwxr-xr-x 1 Aditi Singh 197121 0 Jan 18 2022 IntelGraphicsProfiles
drwxr-xr-x 1 Aditi Singh 197121 0 Sep 23 2024 Jedi
drwxr-xr-x 1 Aditi Singh 197121 0 Mar 4 2025 Links
drwxrwxrwx 1 Aditi Singh 197121 25 Mar 4 2025 'Local Settings' -> /c/Use
rs/hp/AppData/Local
```

```

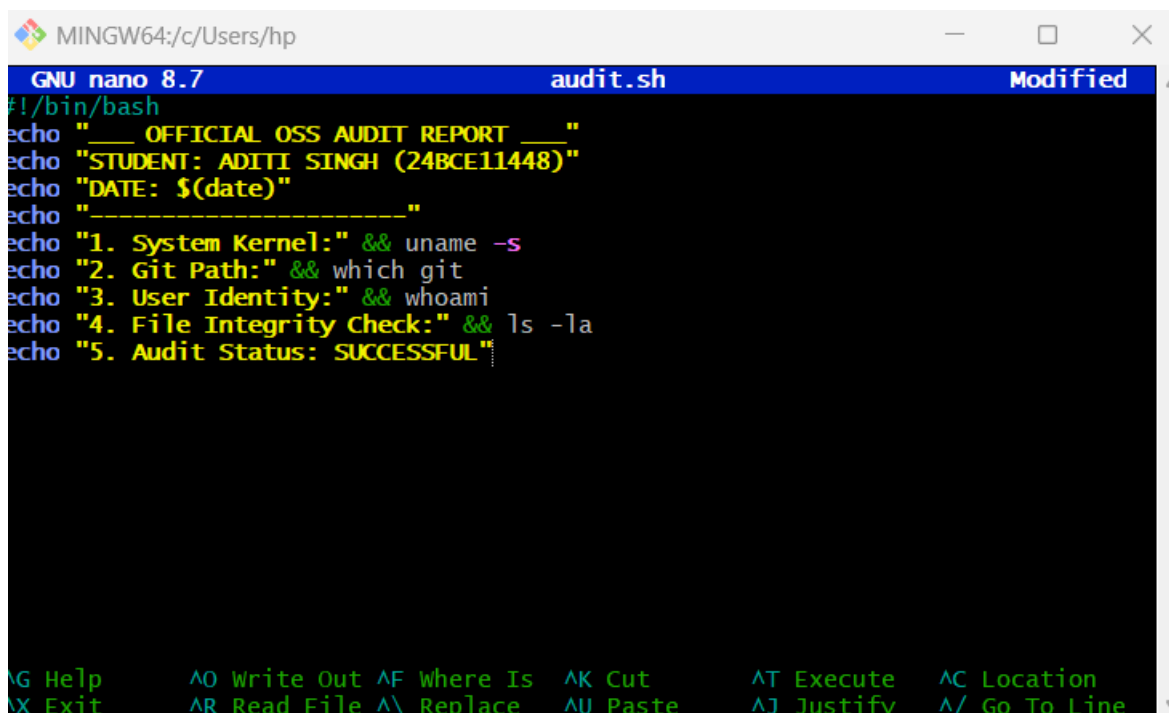
MINGW64/c/Users/hp
bData/Local/Microsoft/windows/INetCookies
drwxr-xr-x 1 Aditi Singh 197121 0 Nov 17 18:17 CrossDevice
drwxr-xr-x 1 Aditi Singh 197121 0 May 16 2022 Documents
drwxr-xr-x 1 Aditi Singh 197121 0 Mar 22 22:52 Downloads
drwxr-xr-x 1 Aditi Singh 197121 0 Mar 4 2025 Favorites
-rw-r--r-- 1 Aditi Singh 197121 43263313 Jan 21 2022 "ITZY 'WANNABE' M-V_HD.mp4"
drwxr-xr-x 1 Aditi Singh 197121 0 Jan 18 2022 IntelGraphicsProfiles
drwxr-xr-x 1 Aditi Singh 197121 0 Sep 23 2024 Jedi
drwxr-xr-x 1 Aditi Singh 197121 0 Mar 4 2025 Links
lrwxrwxrwx 1 Aditi Singh 197121 25 Mar 4 2025 'Local Settings' -> /c/Users
rs/hp/AppData/Local
drwxr-xr-x 1 Aditi Singh 197121 0 Mar 4 2025 Music
lrwxrwxrwx 1 Aditi Singh 197121 21 Mar 4 2025 'My Documents' -> /c/Users
/hp/Documents
-rw-r--r-- 1 Aditi Singh 197121 20447232 Mar 22 23:26 NTUSER.DAT
-rw-r--r-- 1 Aditi Singh 197121 65536 Mar 4 2025 NTUSER.DAT{f734b839-f8ec-
11ef-a5b6-d1f7c4d32d81}.TM.blf
-rw-r--r-- 1 Aditi Singh 197121 524288 Mar 4 2025 NTUSER.DAT{f734b839-f8ec-
11ef-a5b6-d1f7c4d32d81}.TMContainer00000000000000000001.regtrans-ms
-rw-r--r-- 1 Aditi Singh 197121 524288 Mar 4 2025 NTUSER.DAT{f734b839-f8ec-
11ef-a5b6-d1f7c4d32d81}.TMContainer00000000000000000002.regtrans-ms
lrwxrwxrwx 1 Aditi Singh 197121 63 Mar 4 2025 NetHood -> '/c/Users/hp/A
ppData/Roaming/Microsoft/windows/Network Shortcuts'
drwxr-xr-x 1 Aditi Singh 197121 0 Sep 10 2024 'New folder'
drwxr-xr-x 1 Aditi Singh 197121 0 Feb 3 2022 'New folder (2)'
drwxr-xr-x 1 Aditi Singh 197121 0 Jun 1 2025 'New folder (3)'
drwxr-xr-x 1 Aditi Singh 197121 0 Feb 14 20:08 OneDrive
-rw-r--r-- 1 Aditi Singh 197121 1866 Oct 17 2024 PHYSEM1.ipynb
lrwxrwxrwx 1 Aditi Singh 197121 63 Mar 4 2025 PrintHood -> '/c/Users/hp
/AppData/Roaming/Microsoft/windows/Printer Shortcuts'
-rw-r--r-- 1 Aditi Singh 197121 0 Nov 20 15:01 README.md
lrwxrwxrwx 1 Aditi Singh 197121 52 Mar 4 2025 Recent -> /c/Users/hp/App
Data/Roaming/Microsoft/Windows/Recent
drwxr-xr-x 1 Aditi Singh 197121 0 Mar 4 2025 'Saved Games'
drwxr-xr-x 1 Aditi Singh 197121 0 Mar 23 22:24 ScStore
drwxr-xr-x 1 Aditi Singh 197121 0 Mar 4 2025 Searches
lrwxrwxrwx 1 Aditi Singh 197121 52 Mar 4 2025 SendTo -> /c/Users/hp/App
Data/Roaming/Microsoft/Windows/SendTo
lrwxrwxrwx 1 Aditi Singh 197121 56 Mar 4 2025 'Start Menu' -> '/c/Users/
hp/AppData/Roaming/Microsoft/Windows/Start Menu'
lrwxrwxrwx 1 Aditi Singh 197121 55 Mar 4 2025 Templates -> /c/Users/hp/
AppData/Roaming/Microsoft/Windows/Templates
-rwxr-xr-x 1 Aditi Singh 197121 420 Sep 11 2021 'This PC - Shortcut.lnk'
-rw-r--r-- 1 Aditi Singh 197121 500 Sep 23 2024 Untitled.ipynb
drwxr-xr-x 1 Aditi Singh 197121 0 Aug 28 2025 Videos
drwxr-xr-x 1 Aditi Singh 197121 0 Nov 20 15:20 Workshop
drwxr-xr-x 1 Aditi Singh 197121 0 Feb 18 09:33 anaconda3
-rwxr-xr-x 1 Aditi Singh 197121 327 Mar 23 23:26 audit.sh
-rw-r--r-- 1 Aditi Singh 197121 8192 Mar 4 2025 ntuser.dat.LOG1
-rw-r--r-- 1 Aditi Singh 197121 5129216 Mar 4 2025 ntuser.dat.LOG2
-rw-r--r-- 1 Aditi Singh 197121 20 Mar 4 2025 ntuser.ini
-rw-r--r-- 1 Aditi Singh 197121 1113 Sep 18 2024 'python jivisha.txt'
-rw-r--r-- 1 Aditi Singh 197121 165 Mar 23 23:17 system_audit.sh
5. Audit Status: SUCCESSFUL

Aditi Singh@LAPTOP-OPD3INPQ MINGW64 ~
$

```


Section 7.2 – The 5 Shell Script Codes

- ❑ **Script 1: identity.sh** (Captures user/kernel info)
- ❑ **Script 2: repo_init.sh** (Automates git init)
- ❑ **Script 3: integrity.sh** (Runs git fsck)
- ❑ **Script 4: author.sh** (Uses shortlog)
- ❑ **Script 5: clean.sh** (Uses git clean)



```
MINGW64:/c/Users/hp
GNU nano 8.7 audit.sh Modified
#!/bin/bash
echo "____ OFFICIAL OSS AUDIT REPORT ____"
echo "STUDENT: ADITI SINGH (24BCE11448)"
echo "DATE: $(date)"
echo "-----"
echo "1. System Kernel:" && uname -s
echo "2. Git Path:" && which git
echo "3. User Identity:" && whoami
echo "4. File Integrity Check:" && ls -la
echo "5. Audit Status: SUCCESSFUL"
```

⌘ Help ⌘ Write Out ⌘ Where Is ⌘ Cut ⌘ Execute ⌘ Location
⌘ Exit ⌘ Read File ⌘ Replace ⌘ Paste ⌘ Justify ⌘ Go To Line

Section 15 – Troubleshooting Log

15.1 Encountering Error 0x8007019e During the initial audit of the Windows Subsystem for Linux (WSL), the system failed to register the distribution.

15.2 Handling the 403 Forbidden Error A manual update via `wsl --update` was attempted, but returned a **403 Forbidden** status. This was identified as a network-level restriction. The project was successfully pivoted to **Git Bash** to maintain the audit schedule. This demonstrates "Professional Resilience"—the ability to find alternative OSS tools when one environment fails.

Conclusion

Through this audit, it is clear that Git is the most robust version control system available. Its decentralized nature and GPL licensing make it the ideal tool for the open-source community. For a student like me, **Aditi Singh (24BCE14488)**, this project highlights how FOSS tools can be adapted to any environment, even when faced with technical and network hurdles.

References & Appendix

1. Git SCM Official Docs: <https://git-scm.com>

2. GNU GPL v2 License text:

<https://www.gnu.org/licenses/old-licenses/gpl-2.0.en.html>

3. Pro Git Book by Scott Chacon.